

CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 12: Surface Areas and Volumes

Every formula and result of the chapter on one sheet

1	Cuboid	$TSA = 2(lb + bh + hl)$; volume = lbh ; diagonal = $\sqrt{l^2 + b^2 + h^2}$
2	Cube	$TSA = 6a^2$; volume = a^3 ; diagonal = $a\sqrt{3}$
3	Cylinder: curved surface area	$2\pi rh$
4	Cylinder: total surface area	$2\pi r(r + h)$
5	Cylinder: volume	$\pi r^2 h$
6	Cone: slant height	$l = \sqrt{r^2 + h^2}$
7	Cone: curved surface area	πrl
8	Cone: total surface area	$\pi r(l + r)$
9	Cone: volume	$\frac{1}{3}\pi r^2 h$
10	Sphere	Surface area = $4\pi r^2$; volume = $\frac{4}{3}\pi r^3$
11	Hemisphere	$CSA = 2\pi r^2$; $TSA = 3\pi r^2$; volume = $\frac{2}{3}\pi r^3$
12	Cone on a hemisphere	Surface area = $\pi rl + 2\pi r^2$; volume = $\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$
13	Cylinder with two hemispherical ends	Surface area = $2\pi rh + 4\pi r^2$; volume = $\pi r^2 h + \frac{4}{3}\pi r^3$
14	Hollow cylinder	Volume of material = $\pi(R^2 - r^2)h$
15	Units	1 litre = 1000 cm ³ ; 1 m ³ = 1000 litres